

# Probing the Functionality of LiFSI Structural Derivatives as Additives for Li Metal Anodes

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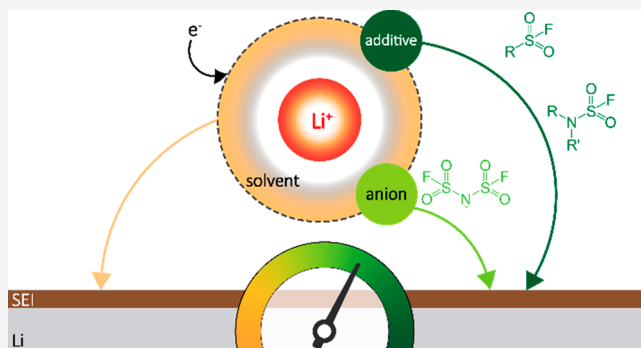


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**ABSTRACT:** Lithium (Li) metal is a compelling replacement for graphite anodes in Li-ion batteries to increase gravimetric energy if the cyclability can be improved. Motivated by high Li Coulombic efficiency (CE) achieved with electrolytes featuring the bis(fluorosulfonyl)imide (FSI<sup>−</sup>) anion, this work examined chemically related sulfonyl/sulfamoyl fluoride additives to correlate FSI<sup>−</sup>-relevant structural features with CE. Across three exemplary carbonate- and glyme-based electrolytes, extended solid, liquid, and gas phase characterizations reveal that Li<sup>+</sup> coordination is necessary yet insufficient for FSI<sup>−</sup> derivatives to affect cycling. Beyond coordination, the reactivity of the baseline solvent and the key structural features of the additives are shown to strongly regulate CE, with possession of an N center—common to sulfamoylfluorides and FSI<sup>−</sup>—consistently leading to higher CE. Some derivatives outperform FSI<sup>−</sup> in short-term cycling; however, they have difficulty competing with the longevity of FSI<sup>−</sup>. These results provide insights for developing improved additives in the future through careful consideration of reactant structure and solvent codesign.



Lithium (Li) metal has the lowest redox potential among all anodes (−3.04 V versus SHE) and an exceptional capacity of 3860 mAh/g, making replacement of graphite with Li one of the leading strategies to significantly boost the energy density of Li-ion batteries.<sup>1,2</sup> However, the Li anode still faces challenges,<sup>3,4</sup> primarily an insufficient Coulombic efficiency (CE), which must reach >99.95% to deliver >1000 cycle life for electric vehicles.<sup>5</sup> The gap between the current and the needed performance is attributed to challenges at the Li solid electrolyte interphase (SEI),<sup>6,7</sup> which when unstable leads to heterogeneous plating, loss of active Li,<sup>8</sup> and/or safety issues in the cell.<sup>9</sup>

Faced with these challenges, researchers have sought improvements in electrolyte design. Among various strategies, including switching from carbonate solvents (CE ~80–90%)<sup>10,11</sup> to less-detrimental ones such as nonfluorinated (~95%)<sup>12,13</sup> and fluorinated (98–99.5%)<sup>14,15</sup> ethers, much focus has been placed on promoting the competitive reduction of salt anions to generate a more compact, uniform, and inorganic SEI.<sup>16</sup> In particular, the bis(fluorosulfonyl)imide anion (FSI<sup>−</sup>, Figure 1a) is the leading candidate, enabling CEs with record values (>99%)<sup>17</sup> when it is used in highly concentrated electrolytes (HCEs, ≥4 M salt)<sup>18–21</sup> that promote favorable FSI<sup>−</sup> coordination to Li<sup>+</sup> and participation in SEI formation. To curb reliance on high salt concentrations

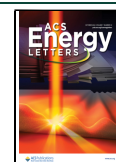
in HCEs, which have high viscosities and costs, local high-concentration electrolytes (LHCEs, 1–2.5 M)<sup>22–24</sup> have been developed that decrease salt concentration yet retain favorable anion-paired configurations around Li<sup>+</sup>. As an alternative strategy, the FSI-like *N,N*-dimethylsulfamoylfluoride solvent (MeFSA, Figure 1a) has been studied, with 2.5 M LiFSI in MeFSA yielding a high CE (>99%) over 150 cycles.<sup>25</sup> The attractive cycling performance was attributed to enhanced reduction of the −SO<sub>2</sub>F group to form LiF, Li<sub>2</sub>O, and other inorganic phases.<sup>17,25,26</sup> Overall, the structural and chemical features responsible for the leading performance of sulfonylimide moieties require a better understanding if electrolytes are to continue to progress beyond FSI<sup>−</sup>, which has already been quite extensively tested in most known solvents and concentrations.

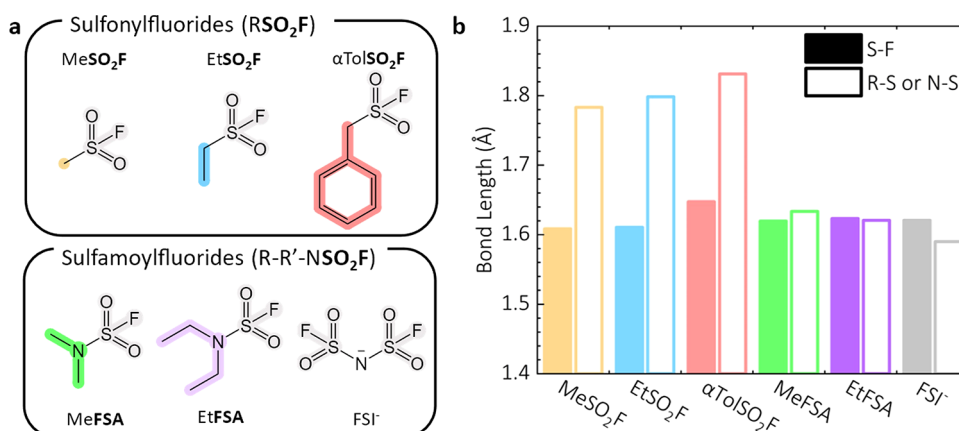
In this context, this study examines a representative set of structural derivatives of FSI<sup>−</sup> that contain a −SO<sub>2</sub>F or −NSO<sub>2</sub>F functionality, with diverse extended structures. To

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**Figure 1.** (a) Chemical structures of the investigated additives and the FSI<sup>-</sup> anion. (b) DFT-calculated S–F (solid bars) and R–S or N–S (open bars) bond lengths of sulfonylfluorides and sulfamoylfluorides.

enable cross-comparison, the reactants are examined at concentrations (3 wt %) typical of additives, given the inability of all structures to dissolve the salt in neat form (additionally, α-TolSO<sub>2</sub>F is a solid). While there is a long history of examining additives for both graphite and Li anodes,<sup>27–35</sup> those having a close structural similarity to LiFSI have received little attention. To provide a systematic investigation in multiple electrolyte environments, the reactants are examined in baseline electrolytes that contain LiFSI (in fluoroethylene carbonate (FEC) and 1,2-dimethoxyethane (DME)), as well as in a non-LiFSI-based electrolyte, Li hexafluorophosphate (LiPF<sub>6</sub>) in ethylene carbonate (EC)/diethyl carbonate (DEC) (LP40).

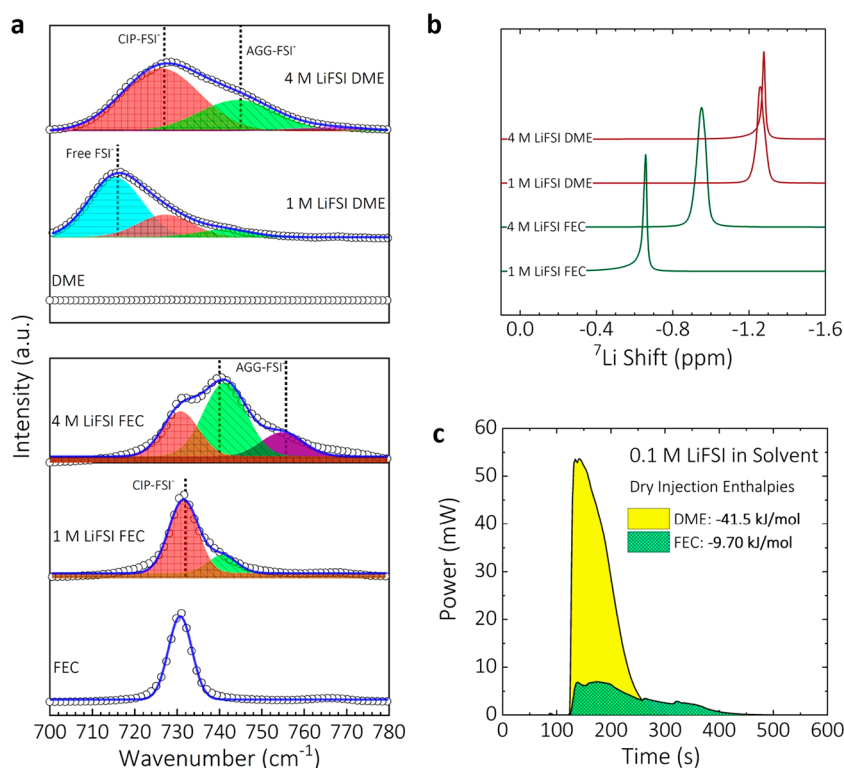
Investigated structures are shown in Figure 1a alongside FSI<sup>-</sup>. Like FSI<sup>-</sup>, the reactants comprise sulfonyl/sulfamoylfluorides, all featuring the –SO<sub>2</sub>F group. Unlike FSI<sup>-</sup>, however, the reactants are neutrally charged and bear only a single –SO<sub>2</sub>F, compared to the two symmetrically bonded to nitrogen (N) in FSI<sup>-</sup>. The sulfonylfluorides (R–SO<sub>2</sub>F), lacking N centers, are more structurally distant from FSI<sup>-</sup>. These include methane sulfonylfluoride (CH<sub>3</sub>SO<sub>2</sub>F, MeSO<sub>2</sub>F), ethane sulfonylfluoride (C<sub>2</sub>H<sub>5</sub>SO<sub>2</sub>F, EtSO<sub>2</sub>F), and α-toluene sulfonylfluoride (C<sub>6</sub>H<sub>4</sub>CH<sub>2</sub>SO<sub>2</sub>F, α-TolSO<sub>2</sub>F). In contrast, the sulfamoylfluorides (R,R'–NSO<sub>2</sub>F) contain N and are therefore structurally closer to FSI<sup>-</sup>, and include N,N-dimethyl sulfamoylfluoride ((CH<sub>3</sub>)<sub>2</sub>NSO<sub>2</sub>F, MeFSA) and N,N-diethyl sulfamoylfluoride ((C<sub>2</sub>H<sub>5</sub>)<sub>2</sub>NSO<sub>2</sub>F, EtFSA). Across both classes, the R groups were chosen to prioritize structural simplicity and to ensure adequate solubility in the baseline electrolytes of interest; for example, alternatives including perfluorinated sulfonylfluorides were insoluble in ether and carbonate solvents. In addition, the selected additives are readily commercially accessible and generally safe for laboratory use (Table S1).

Like FSI<sup>-</sup>, the molecular structures possess multiple possible sites for fragmentation upon participation in SEI formation. Hence, density functional theory (DFT) calculations were conducted to compare likely sites for reactivity. While all structures, including FSI<sup>-</sup>, possess similar S–F bond lengths (~1.60–1.65 Å; Figure 1b), significant variability in R–S or N–S bond lengths was observed: those in sulfonylfluorides (R–S bonds, 1.78–1.83 Å) are notably longer than those in sulfamoylfluorides (N–S bonds, 1.59–1.63 Å). This suggests that, without a central N atom, the cleavage of R–SO<sub>2</sub>F is more facile. On the other hand, the sulfamoylfluorides have

comparable N–S and S–F bond lengths, indicating less-preferential sites for fragmentation. The sulfamoylfluorides are also more similar to FSI<sup>-</sup> in terms of bond lengths. Implications of these observations on the reactivity, SEI chemistry, and CE are discussed later.

Raman spectroscopy, <sup>7</sup>Li nuclear magnetic resonance (NMR) spectroscopy and reaction calorimetry were first conducted to understand the coordination environment of Li<sup>+</sup> in the baseline FEC and DME electrolytes with LiFSI salt prior to additive incorporation. Two salt concentrations were examined for each solvent to provide better accuracy in interpreting the Raman peak assignments and NMR shielding trends: 1 and 4 M LiFSI in FEC<sup>17</sup> and 1 and 4 M LiFSI in DME.<sup>19</sup> In Raman, four FSI<sup>-</sup> bands are distinguishable across all electrolytes (Figure 2a), with uncoordinated/free (cyan) FSI<sup>-</sup> at 716 cm<sup>-1</sup> and contact-ion paired (CIP, red) and higher-order aggregates (AGG, green and magenta) at 732 and 741–756 cm<sup>-1</sup>, respectively. Starting with 1 M LiFSI FEC, a Raman-active solvent mode at 730 cm<sup>-1</sup> is convoluted with the CIP–FSI<sup>-</sup> peak; nevertheless, an insignificant peak intensity at 716 cm<sup>-1</sup> indicates negligible population of free FSI<sup>-</sup>, with AGG structures only becoming significant at 4 M. The corresponding NMR data (Figure 2b) revealed strong upfield chemical shifts (from –0.66 to –0.95 ppm) occurring as the LiFSI concentration was increased, indicating that FSI<sup>-</sup> anions shield Li<sup>+</sup> substantially more strongly than FEC. We confirmed the tendency toward low degrees of salt dissociation in FEC using solid injection microcalorimetry, which measures the heat of dissolution upon injecting a fixed amount of salt into solvent. Even at the relatively dilute 0.1 M suitable for the measurement (Figure 2c), FEC exhibited only a weakly exothermic peak of –9.70 kJ/mol<sub>Li+</sub>, indicating low degrees of salt dissociation with weak competitive solvation.

In contrast, DME promoted a heightened proportion of free FSI<sup>-</sup> at 1 M (Figure 2a), whereas at 4 M, the majority of FSI<sup>-</sup> were in CIP and AGG. In NMR, the Li<sup>+</sup> peak in 1 M LiFSI DME (–1.26 ppm) was already located the farthest upfield and exhibited negligible further change as the salt concentration was increased, despite evident increases in ion pairing from Raman (Figure S1 shows a wider concentration range). This indicates that the degree of Li<sup>+</sup> shielding by FSI<sup>-</sup> is more comparable to that of DME; both FSI<sup>-</sup> and DME shield Li<sup>+</sup> more effectively than FEC. DME also exhibited a more strongly exothermic dissolution heat of –41.5 kJ/mol<sub>Li+</sub> at 0.1 M compared to FEC, reflecting a larger degree of Li<sup>+</sup>–FSI<sup>-</sup>



**Figure 2.** Characterization of baseline electrolytes without additives. (a) Raman spectroscopy of the  $\text{FSI}^-$  band for neat DME or DME with 1 or 4 M LiFSI and for neat FEC or FEC with 1 or 4 M LiFSI. “CIP” refers to contact-ion pairs; “AGG” denotes aggregates and higher-order clusters. (b)  $^7\text{Li}$  NMR spectra showing the  $\text{Li}^+$  chemical shift in each baseline electrolyte as a function of the examined salt concentrations. Chemical shifts were externally referenced to 1 M LiCl  $\text{D}_2\text{O}$ . (c) Power profile upon dry injection of 0.1 M LiFSI salt into neat DME or FEC solvent, measured using reaction microcalorimetry. Molar heats of dissolution ( $\text{kJ/mol}_{\text{Li}^+}$ ) are calculated by integrating the power profiles with respect to time and normalizing by the salt quantity, with the convention that negative values correspond to net exothermic processes.

dissociation (endothermic) and  $\text{Li}^+$ –solvent association (strongly exothermic), attributed to the ability of DME to chelate  $\text{Li}^+$ .<sup>36,37</sup> Comparable trends in NMR shifts were observed for the same 0.1 M LiFSI concentrations used for microcalorimetry (Figure S2). The implications of these trends in Figure 2 on the ability of  $-\text{SO}_2\text{F}$  and  $-\text{NSO}_2\text{F}$  structures to coordinate  $\text{Li}^+$  are discussed later.

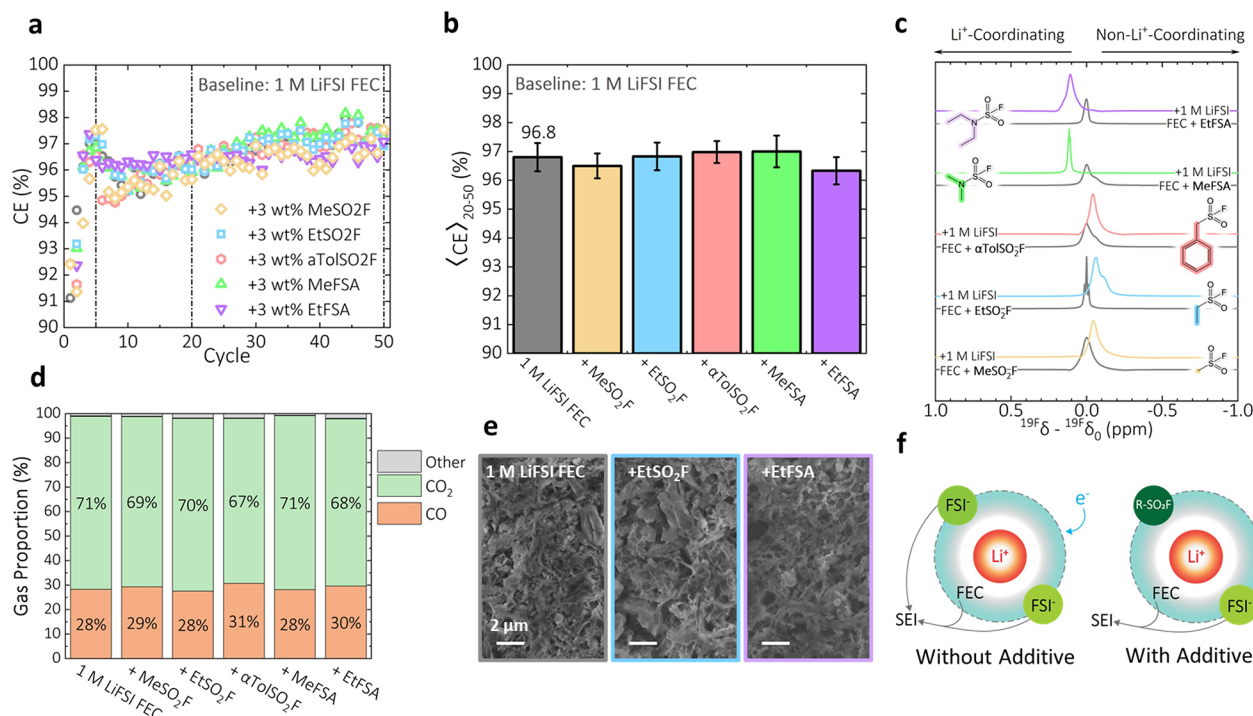
Li cycling was first studied in 1 M LiFSI in FEC, containing negligible free  $\text{FSI}^-$  anions. Figure 3a shows CEs of cells cycled at  $0.25 \text{ mA/cm}^2$  for  $0.25 \text{ mAh/cm}^2$  over 50 cycles without and with 3 wt % of additives ( $\sim 0.3\text{--}0.5 \text{ M}$ , Table S1). The initial baseline CE (average 96.2%, cycles 5–20, Figure S3a) agrees well with the literature (96.1%, 100 cycles).<sup>17</sup> All cells exhibited a steep CE increase from  $\sim 91\text{--}93\%$  (cycle 1) to  $\sim 96\text{--}98\%$  over the first 10 cycles, after which the CE increased more gradually. Averaged CEs over cycles 20–50 are shown in Figure S3b, and complete tabulated values are given in Table S2, indicating that the variation of CE across additives at a given cycle is negligible. The CEs of additional trials are plotted in Figure S4 and showed good reproducibility.

NMR was further used to investigate whether and how additives affect the  $\text{Li}^+$  coordination environment in FEC. Given that the additives have lower concentrations in comparison to the 1 M salt, and  $^7\text{Li}$  shifts reflect multiple complex interactions (Figure S5a); better sensitivity is achieved using  $^{19}\text{F}$  NMR, as each additive has a unique sulfonylfluoride peak (Figure S6). Figure 3c shows these peaks relative to their position in neat solvent. In all cases a shift is observed when 1 M LiFSI is added. While the sulfonylfluorides ( $\text{MeSO}_2\text{F}$ ,  $\text{EtSO}_2\text{F}$ , and  $\alpha\text{TolSO}_2\text{F}$ ) exhibited small ( $<0.1$

ppm) upfield shifts indicative of slightly increased shielding, the sulfonylfluorides ( $\text{MeFSa}$  and  $\text{EtFSa}$ ) exhibited downfield shifts ( $\sim 0.1 \text{ ppm}$ ). We interpret these in the same manner as  $\text{FSI}^-$  shielding/deshielding (Figure S7), i.e., with deshielding indicating stronger  $\text{Li}^+$  coordination.<sup>15</sup> Consistently, upfield shifts have been attributed elsewhere to fluorine-shielding self-interactions, such as hydrogen bonding, among bulk (uncoordinated) molecules that are enhanced when the indicated molecules are driven out of the coordination environment.<sup>38</sup> The results suggest that sulfonylfluorides coordinate weakly to  $\text{Li}^+$ , while sulfonylfluorides do not. Regardless, even the species that do coordinate to  $\text{Li}^+$  have a curiously negligible effect on CE.

To examine this more closely, the evolved gases from Li–electrolyte reactions were extracted from the cell headspace for gas chromatography (GC) analysis and were found to be similar for all additives (Figure 3d and Figure S8a;  $\text{H}_2$  evolution discussed in Table S3). The major gases were  $\text{CO}$  and  $\text{CO}_2$ , identical with those formed in the baseline electrolyte and attributed to FEC reactivity,<sup>39</sup> as additives do not contain a  $\text{CO/CO}_2$ -evolving functionality. SEI compositions, determined by XPS analysis (Figure S9), were also similar in all cases. Specifically, F 1s spectra show that  $\text{LiF}$ , followed by  $-\text{SO}_2\text{F}$ , are the major components and their intensities and relative proportions are electrolyte-invariant. The S 2p spectra also show little variation in  $-\text{SO}_2\text{F}$  and  $-\text{S}_2\text{O}_4$ ; C 1s spectra indicate similar  $-\text{CO}_3$  and  $-\text{CO}$  contents, and N 1s and O 1s spectra are also similar. We attribute the variation in C–C peak intensity in the C 1s spectra to adventitious carbon introduced to sample surfaces





**Figure 3.** Galvanostatic cycling performance (current 0.25 mA/cm<sup>2</sup> and capacity 0.25 mAh/cm<sup>2</sup>) and characterization of Li–Cu cells for the 1 M LiFSI FEC electrolyte. (a) Variation of CE with cycle number. (b) CE averaged over the 20th to 50th cycles. Error bars represent the variation across the specified cycles for a single cell. (c) <sup>19</sup>F NMR spectra of 3 wt % additives in FEC without salt and with 1 M LiFSI. Chemical shifts were referenced to the –SO<sub>2</sub>F fluorine of the respective additive in neat FEC at 0 ppm. (d) Gas chromatography analysis of the cell headspace following 30 plating/stripping cycles. “Other” refers to the combined proportions of C<sub>2</sub>H<sub>6</sub>, C<sub>2</sub>H<sub>4</sub>, CH<sub>4</sub>, and C<sub>2</sub>H<sub>2</sub>. (e) SEM images of the morphologies of Li plated on Cu substrates for representative additives after 20 cycles. (f) Schematic of proposed coordinating and SEI-building species in LiFSI FEC.

during their transfer and measurement. X-ray absorption near edge spectroscopy (XANES) (Figure S10) further confirmed that the SEI surface (TEY) and bulk (FLY) are predominantly LiF, and these were similar for all electrolytes. Finally, SEM imaging (Figure 3e and Figure S3b) of Li deposits after 20 cycles indicated that the deposition morphologies were comparably mossy and porous across the samples, with no discernible differences.

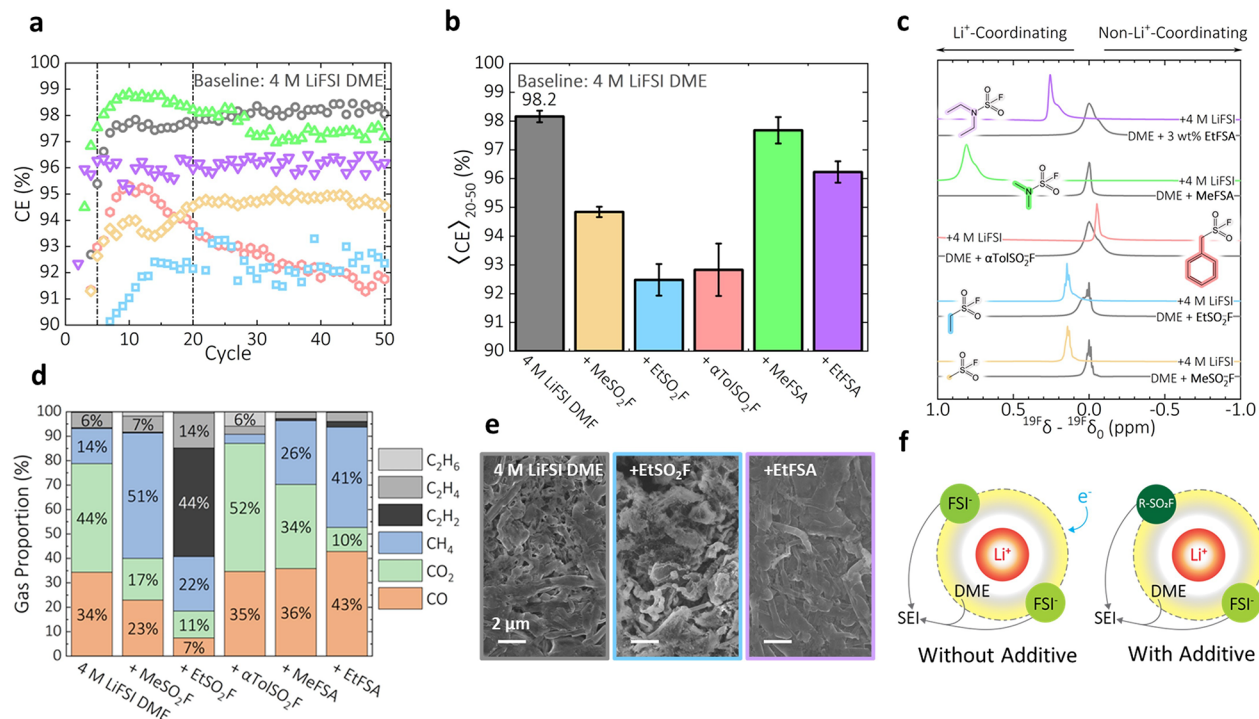
Collectively, these data unambiguously demonstrate that the SEI is formed from species common to all systems (Figure 3f), namely, FSI<sup>-</sup> and FEC, even in cases where additives do coordinate. Hence, FSI<sup>-</sup> and FEC are kinetically selected over other additives in the coordination environment of Li<sup>+</sup>, even though FEC is relatively weakly solvating. The implied activity of FEC is consistent with previous studies which have shown that the fluorination of FEC imparts a lower LUMO energy and elevated reactivity in the bulk solvent compared to other carbonates,<sup>40</sup> and that once Li<sup>+</sup> is coordinated by even a single FEC in 1.1 M LiPF<sub>6</sub> in ethyl methyl carbonate (EMC), FEC dominates the performance and CE is markedly improved.<sup>41</sup>

The above observations continue to hold true for a higher salt concentration of 4 M LiFSI in FEC (Figure S11), which has a higher baseline CE of 97.6%. This CE remained unchanged upon incorporation of additives, even though the <sup>19</sup>F NMR indicated an even stronger tendency toward Li<sup>+</sup> coordination for the sulfamoylfluorides (Figure S12). Increasing the salt concentration therefore appears to primarily elevate FSI<sup>-</sup> reduction to yield higher average CEs.

Similar characterizations were next conducted in 4 M LiFSI in DME. This higher concentration was examined given that

lower concentrations exhibited poor cyclability due to the use of Whatman separators (see Experimental Methods in the Supporting Information). Figure 4a,b and Figure S13a show that the baseline electrolyte CE over cycles 5–20 was 97.2% and increased to 98.2% over cycles 20–50.<sup>19</sup> Reproduced cells are shown in Figure S14. Remarkably, unlike FEC, the cycling performance in DME was highly sensitive to the additive structure: CE ranged from 91.6% (EtSO<sub>2</sub>F) to 98.1% (MeFSA) over cycles 5–20, with the lowest CE consistently occurring with the sulfonylfluorides and the highest (comparable to the baseline) with the sulfamoylfluorides (tabulated data in Table S4). The <sup>19</sup>F NMR peaks of all additives (Figure 4c; <sup>7</sup>Li spectra in Figure S5b and unshifted <sup>19</sup>F spectra in Figure S15) exhibited downfield shifts upon adding 4 M LiFSI to 3 wt % additive in DME, with the exception of αTolSO<sub>2</sub>F, whose anomalous shift may be due to cation–π interactions involving its aromatic ring.<sup>42</sup> This shows that nearly all additives coordinate with Li<sup>+</sup> in DME. Consistent with observations in FEC, MeFSA exhibits the strongest interaction with Li<sup>+</sup> in DME, followed by EtFSA.

Given the disparity of cycling outcomes in 4 M LiFSI in DME, it is informative to investigate the SEI and gas compositions for each electrolyte in turn. In the baseline electrolyte, GC analysis (Figure 4d and Figure S8b) showed primarily CO<sub>2</sub>, CO, CH<sub>4</sub>, and minor amounts of C<sub>2</sub> gases (e.g., C<sub>2</sub>H<sub>4</sub>, C<sub>2</sub>H<sub>2</sub>) from DME reduction. In XPS (Figure S16), comparable –SO<sub>2</sub>F and LiF intensities in the F 1s high-resolution spectra, and –SO<sub>2</sub>F/S<sub>2</sub>O<sub>4</sub> features in the S 2p spectra are attributed to FSI<sup>-</sup> reduction. The C 1s spectrum showed significant –CO solvent-derived features from DME.



**Figure 4.** Galvanostatic cycling performance (current 0.25 mA/cm<sup>2</sup> and capacity 0.25 mAh/cm<sup>2</sup>) and characterization of Li–Cu cells for the 4 M LiFSI DME electrolyte. (a) Variation of CE with cycle number; colors representing electrolytes are the same as in (b, c). (b) CE averaged over the 20th to 50th cycles. Error bars represent the variation across the specified cycles for a single trial. (c) <sup>19</sup>F NMR spectra of 3 wt % additives in DME without salt and with 4 M LiFSI. (d) Headspace gas byproducts accumulated over 30 plating/stripping cycles as measured by gas chromatography. (e) SEM images of the morphologies of Li plated on Cu substrates for representative additives after 20 cycles. (f) Schematic of proposed coordinating and SEI-building species in LiFSI DME.

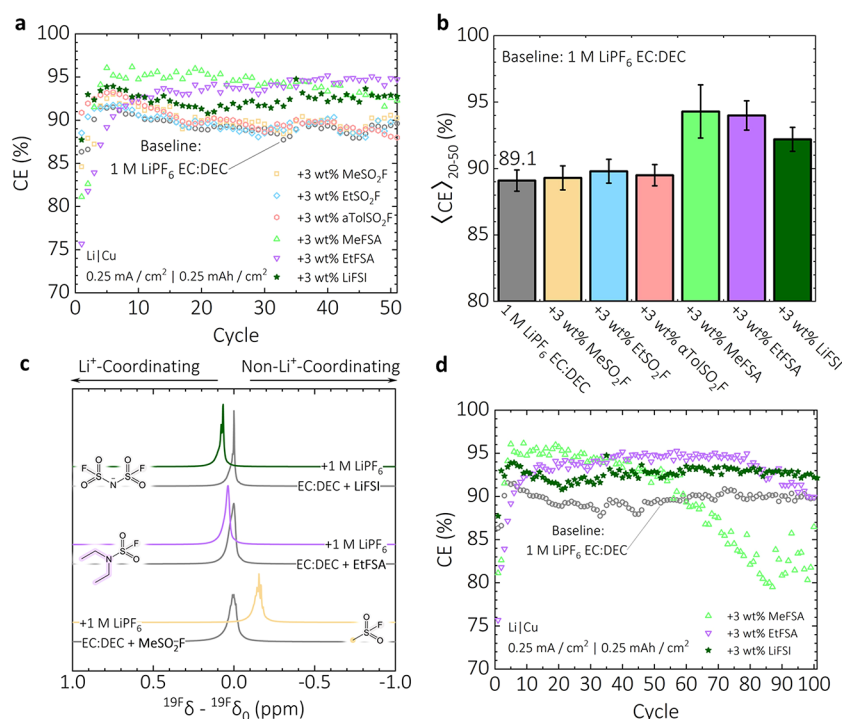
With MeSO<sub>2</sub>F, GC measurements showed increased CH<sub>4</sub> evolution consistent with cleavage of the lengthened R–S bond, as shown in Figure 1b. The XPS F 1s and S 2p spectra indicated proportionally increased LiF with respect to –SO<sub>2</sub>F and enriched –S<sub>2</sub>O<sub>4</sub> compared to –SO<sub>2</sub>F, respectively, versus the baseline. These data indicate that MeSO<sub>2</sub>F is fragmented in the SEI in a manner different from FSI<sup>–</sup> to yield unique proportions of constituent LiF and sulfates. Hence, MeSO<sub>2</sub>F does participate in SEI formation, but not beneficially, given that the average CE over cycles 20–50 decreased from 98.2% to 94.8%. The other two sulfonylfluorides, EtSO<sub>2</sub>F and αTolSO<sub>2</sub>F, also formed heightened LiF with respect to –SO<sub>2</sub>F and had decreased CEs (92.5% and 92.8%, respectively). EtSO<sub>2</sub>F generated enhanced C<sub>2</sub> gas products from the ethyl tail group, whereas αTolSO<sub>2</sub>F did not evolve significant additive-derived gas products, attributed to the high molecular weight of the toluene-derived side group. As a result, the gas signature with αTolSO<sub>2</sub>F was most similar to that of the baseline electrolyte.

In contrast, the sulfamoylfluoride additives yielded SEI compositions, CEs, and plating morphologies more similar to those of 4 M LiFSI DME. XPS F 1s showed more balanced –SO<sub>2</sub>F and LiF contributions, with lower LiF proportions compared to the sulfonylfluorides. The S 2p spectra show that –SO<sub>2</sub>F is consistently more prominent than –S<sub>2</sub>O<sub>4</sub> with sulfamoylfluorides. Consequently, the degree of fragmentation within –SO<sub>2</sub>F appears to be less extensive in some sulfamoylfluoride additives than in sulfonylfluoride additives, although EtSO<sub>2</sub>F appears to present an exception to this rule. XANES spectra (Figure S17) showed, in agreement with XPS, that sulfonylfluoride additives led to enhanced LiF in both the

surface and bulk SEI compared to the baseline, while LiF in sulfamoylfluoride additives was not obviously increased. The cycling sensitivity to additives in DME is further supported by the SEM images of plated Li (Figure 4e and Figure S13b), which show that within the first 20 cycles, electrolytes with sulfamoylfluorides led to denser Li deposition than those with sulfonylfluorides.

The collective results show that the sensitivity of CE to the additive R group is substantially higher in DME vs FEC, confirming that additives do contribute to SEI formation in DME, as shown schematically in Figure 4f. We attribute this to the relatively low amount of DME available for coordination (2.4 DME per Li<sup>+</sup><sup>36</sup>) at high salt concentration, allowing additives to coordinate; the comparable Li<sup>+</sup> shielding strengths of FSI<sup>–</sup> and DME from Figure 2, which may provide more numerous sites for additives to displace a coordinating molecule; and the relative cathodic stability of DME,<sup>18,19</sup> allowing additives to kinetically compete once coordinated. Considering the results in DME, two critical functions of the N center of FSI<sup>–</sup> derivative structures become apparent: (1) it consistently supports Li<sup>+</sup> coordination in both solvents examined so far and (2) it helps to avert complete structural fragmentation, consistent with DFT predictions in Figure 1b. These two properties support better cycling performances of sulfamoylfluorides compared to sulfonylfluorides.

We finally sought to examine whether these design guidelines are generalizable beyond the two electrolytes studied so far by testing their performance in a substantially different electrolyte environment, 1 M LiPF<sub>6</sub> in EC:DEC (LP40). This electrolyte system has the lowest baseline CE (average of 89.1%, cycles 20–50) among those studied herein,



**Figure 5.** Cycling performance of sulfonyl/sulfamoyl fluoride and LiFSI additives in 1 M LiPF<sub>6</sub> EC:DEC (LP40) electrolyte. (a) Variation of CE with cycle number. (b) CE averaged over the 20th to 50th cycles. (c) <sup>19</sup>F NMR spectra for MeSO<sub>2</sub>F, EtFSA, and LiFSI additives in LP40. (d) Extended cycling up to 100 cycles of LP40 electrolytes with sulfamoylfluoride additives.

given the known problematic reactivity of carbonates. The design guidelines identified in the preceding paragraph lead to the hypothesis that the sulfamoylfluorides have a higher likelihood to coordinate to Li<sup>+</sup> and thus improve the CE. Indeed, as shown in Figures 5a,b, the sulfonylfluorides had a negligible effect on the CE (~89.1–89.8%, comparable to baseline), whereas the sulfamoylfluorides, as well as additive-quantity (3 wt %) LiFSI, improved it: CEs were 94.3%, 94.0%, and 92.2% for MeFSA, EtFSA, and FSI<sup>−</sup>, respectively, over cycles 20–50 (additional trials are shown in Figure S18). This affirms the previous findings that cycling outcomes are divergent along the line separating sulfonylfluorides and sulfamoylfluorides and can further be directly related to Li<sup>+</sup> coordination: <sup>19</sup>F NMR spectra for a representative sulfonylfluoride (MeSO<sub>2</sub>F), a sulfamoylfluoride (EtFSA), and FSI<sup>−</sup> reveal that the latter two coordinate Li<sup>+</sup>, whereas MeSO<sub>2</sub>F does not (Figure 5c; full spectra are given in Figure S19). Figure S20 further shows that the SEI composition, relative headspace gas proportions, and plating morphology of LP40 were unchanged by the MeSO<sub>2</sub>F additive but were substantially more sensitive to EtFSA and FSI<sup>−</sup>, which enriched −SO<sub>2</sub>F and slightly increased LiF compared to the SEI of LP40.

Despite the improving initial performance, however, Figure 5d shows that the MeFSA CE gradually declined over cycling. EtFSA, which was initially more stable, also exhibited a decline in CE after a longer ~90 cycles. In contrast, with FSI<sup>−</sup>, CE values were sustained for 100 cycles with no decline. Hence, LiFSI can still outperform any additive in terms of longevity, even though its CE is below that of the sulfamoylfluorides initially. These outcomes collectively and finally reveal that N content alone cannot fully explain the high and sustained CE attainable with FSI<sup>−</sup>. The exceptional stability of FSI<sup>−</sup> at extended cycles is potentially due to its absence of detrimental gaseous byproduct generation during cycling or its negative

charge and possession of two −SO<sub>2</sub>F groups that increase the number of available Li<sup>+</sup> coordination sites.<sup>43</sup> Such properties were not captured by any additives studied herein and will be investigated in the future.

Overall, the comprehensive data provide insights into the baseline electrolyte properties under which FSI<sup>−</sup>-like motifs effectively modulate the Li CE. For a desired structure to affect cycling, it is equally important to consider (1) whether the additive is Li<sup>+</sup>-coordinating, and (2) what is the hierarchy of reactivity and the degree of SEI benefit among the solvent, anion, and electrolyte additives. In FEC electrolytes, where FSI<sup>−</sup> and a reactive solvent dominate SEI-forming reactions, additives have neither a positive nor negative impact on CE. With less reactive solvents such as DME, additive reduction can be promoted over solvent reduction and the intrinsic functions of certain structures could be more clearly discerned. In terms of additive sensitivity, LP40 appears to fall in the same category as DME; however, in that instance, a problematically reactive solvent/anion combination can be markedly improved upon by introducing competitive −N−SO<sub>2</sub>F in the coordination sphere of Li<sup>+</sup>. Future efforts into the design of FSI-like solvents and/or additives are recommended to focus on N-containing motifs in greater detail, which may enable more versatility of the R-group design without problematic fragmentation and gas generation. Such additives are expected to have the most significant benefits in (1) systems where strongly coordinated solvents, such as glymes, are under-coordinated, e.g. at higher salt concentrations; or (2) systems with non-FEC carbonates, given the noted difficulty to compete with FEC.



## ■ ASSOCIATED CONTENT

### SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsenerylett.2c01818>.

Experimental methods,  $^7\text{Li}$  NMR spectra of LiFSI in organic solvents, additional SEM images, reproducibility of cells, full  $^7\text{Li}$  NMR and  $^{19}\text{F}$  NMR spectra of baseline electrolyte solutions with additives, absolute non- $\text{H}_2/\text{Ar}$  gas quantities evolved, high-resolution XPS spectra, X-ray near-edge absorption spectra (XANES), 4 M LiFSI FEC cycling data, GC, XPS, and SEM characterization of LP40 electrolytes, additive properties and concentrations, average CE values, and hydrogen gas evolution and water impurities (PDF)

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### Notes

The authors declare no competing financial interest.

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## ■ REFERENCES

- (1) Schmuck, R.; Wagner, R.; Hörpel, G.; Placke, T.; Winter, M. Performance and cost of materials for lithium-based rechargeable automotive batteries. *Nature Energy* **2018**, *3* (4), 267–278.
- (2) Liu, J.; Bao, Z.; Cui, Y.; Dufek, E. J.; Goodenough, J. B.; Khalifah, P.; Li, Q.; Liaw, B. Y.; Liu, P.; Manthiram, A.; et al. Pathways for practical high-energy long-cycling lithium metal batteries. *Nature Energy* **2019**, *4* (3), 180–186.
- (3) Lin, D.; Liu, Y.; Cui, Y. Reviving the lithium metal anode for high-energy batteries. *Nature Nanotechnol.* **2017**, *12* (3), 194–206.
- (4) Xu, W.; Wang, J.; Ding, F.; Chen, X.; Nasybulin, E.; Zhang, Y.; Zhang, J.-G. Lithium metal anodes for rechargeable batteries. *Energy Environ. Sci.* **2014**, *7* (2), 513–537.
- (5) Hobold, G. M.; Lopez, J.; Guo, R.; Minafra, N.; Banerjee, A.; Shirley Meng, Y.; Shao-Horn, Y.; Gallant, B. M. Moving beyond 99.9% Coulombic efficiency for lithium anodes in liquid electrolytes. *Nature Energy* **2021**, *6* (10), 951–960.
- (6) Xu, K. Nonaqueous liquid electrolytes for lithium-based rechargeable batteries. *Chem. Rev.* **2004**, *104* (10), 4303–4418.
- (7) Peled, E.; Menkin, S. SEI: past, present and future. *J. Electrochem. Soc.* **2017**, *164* (7), A1703.
- (8) Chen, K.-H.; Wood, K. N.; Kazyak, E.; LePage, W. S.; Davis, A. L.; Sanchez, A. J.; Dasgupta, N. P. Dead lithium: mass transport effects on voltage, capacity, and failure of lithium metal anodes. *Journal of Materials Chemistry A* **2017**, *5* (23), 11671–11681.
- (9) Tikekar, M. D.; Choudhury, S.; Tu, Z.; Archer, L. A. Design principles for electrolytes and interfaces for stable lithium-metal batteries. *Nature Energy* **2016**, *1* (9), 1–7.
- (10) Selim, R.; Bro, P. Some observations on rechargeable lithium electrodes in a propylene carbonate electrolyte. *J. Electrochem. Soc.* **1974**, *121* (11), 1457.
- (11) Bieker, G.; Winter, M.; Bieker, P. Electrochemical in situ investigations of SEI and dendrite formation on the lithium metal anode. *Phys. Chem. Chem. Phys.* **2015**, *17* (14), 8670–8679.
- (12) Foos, J.; Stolki, T. A new ether solvent for lithium cells. *J. Electrochem. Soc.* **1988**, *135* (11), 2769.
- (13) Koch, V.; Young, J. 2-Methyltetrahydrofuran—lithium hexafluoroarsenate: a superior electrolyte for the secondary lithium electrode. *Science* **1979**, *204* (4392), 499–501.
- (14) Yu, Z.; Wang, H.; Kong, X.; Huang, W.; Tsao, Y.; Mackanic, D. G.; Wang, K.; Wang, X.; Huang, W.; Choudhury, S.; et al. Molecular design for electrolyte solvents enabling energy-dense and long-cycling lithium metal batteries. *Nature Energy* **2020**, *5* (7), 526–533.
- (15) Ma, P.; Mirmira, P.; Amanchukwu, C. V. Effect of building block connectivity and ion solvation on electrochemical stability and ionic conductivity in novel fluoroether electrolytes. *ACS Central Science* **2021**, *7* (7), 1232–1244.
- (16) Yamada, Y.; Wang, J.; Ko, S.; Watanabe, E.; Yamada, A. Advances and issues in developing salt-concentrated battery electrolytes. *Nature Energy* **2019**, *4* (4), 269–280.
- (17) Suo, L.; Xue, W.; Gobet, M.; Greenbaum, S. G.; Wang, C.; Chen, Y.; Yang, W.; Li, Y.; Li, J. Fluorine-donating electrolytes enable highly reversible 5-V-class Li metal batteries. *Proc. Natl. Acad. Sci. U. S. A.* **2018**, *115* (6), 1156–1161.
- (18) Park, M. S.; Ma, S. B.; Lee, D. J.; Im, D.; Doo, S.-G.; Yamamoto, O. A highly reversible lithium metal anode. *Sci. Rep.* **2014**, *4* (1), 1–8.
- (19) Qian, J.; Henderson, W. A.; Xu, W.; Bhattacharya, P.; Engelhard, M.; Borodin, O.; Zhang, J.-G. High rate and stable cycling of lithium metal anode. *Nat. Commun.* **2015**, *6* (1), 1–9.
- (20) Yamada, Y.; Furukawa, K.; Sodeyama, K.; Kikuchi, K.; Yaegashi, M.; Tateyama, Y.; Yamada, A. Unusual stability of acetonitrile-based

superconcentrated electrolytes for fast-charging lithium-ion batteries. *J. Am. Chem. Soc.* **2014**, *136* (13), 5039–5046.

(21) Zeng, Z.; Murugesan, V.; Han, K. S.; Jiang, X.; Cao, Y.; Xiao, L.; Ai, X.; Yang, H.; Zhang, J.-G.; Sushko, M. L.; et al. Non-flammable electrolytes with high salt-to-solvent ratios for Li-ion and Li-metal batteries. *Nature Energy* **2018**, *3* (8), 674–681.

(22) Ren, X.; Zou, L.; Cao, X.; Engelhard, M. H.; Liu, W.; Burton, S. D.; Lee, H.; Niu, C.; Matthews, B. E.; Zhu, Z.; et al. Enabling high-voltage lithium-metal batteries under practical conditions. *Joule* **2019**, *3* (7), 1662–1676.

(23) Ren, X.; Gao, P.; Zou, L.; Jiao, S.; Cao, X.; Zhang, X.; Jia, H.; Engelhard, M. H.; Matthews, B. E.; Wu, H.; et al. Role of inner solvation sheath within salt–solvent complexes in tailoring electrode/electrolyte interphases for lithium metal batteries. *Proc. Natl. Acad. Sci. U. S. A.* **2020**, *117* (46), 28603–28613.

(24) Wu, H.; Jia, H.; Wang, C.; Zhang, J. G.; Xu, W. Recent progress in understanding solid electrolyte interphase on lithium metal anodes. *Adv. Energy Mater.* **2021**, *11* (5), 2003092.

(25) Xue, W.; Shi, Z.; Huang, M.; Feng, S.; Wang, C.; Wang, F.; Lopez, J.; Qiao, B.; Xu, G.; Zhang, W.; et al. FSI-inspired solvent and “full fluorosulfonyl” electrolyte for 4 V class lithium-metal batteries. *Energy Environ. Sci.* **2020**, *13* (1), 212–220.

(26) Shkrob, I. A.; Marin, T. W.; Zhu, Y.; Abraham, D. P. Why bis (fluorosulfonyl) imide is a “magic anion” for electrochemistry. *J. Phys. Chem. C* **2014**, *118* (34), 19661–19671.

(27) Aurbach, D.; Gamolsky, K.; Markovsky, B.; Gofer, Y.; Schmidt, M.; Heider, U. On the use of vinylene carbonate (VC) as an additive to electrolyte solutions for Li-ion batteries. *Electrochimica Acta* **2002**, *47* (9), 1423–1439.

(28) Markevich, E.; Salitra, G.; Fridman, K.; Sharabi, R.; Gershtinsky, G.; Garsuch, A.; Semrau, G.; Schmidt, M. A.; Aurbach, D. Fluoroethylene carbonate as an important component in electrolyte solutions for high-voltage lithium batteries: role of surface chemistry on the cathode. *Langmuir* **2014**, *30* (25), 7414–7424.

(29) Osaka, T.; Momma, T.; Tajima, T.; Matsumoto, Y. Enhancement of lithium anode cyclability in propylene carbonate electrolyte by CO<sub>2</sub> addition and its protective effect against H<sub>2</sub>O impurity. *J. Electrochem. Soc.* **1995**, *142* (4), 1057.

(30) Aurbach, D.; Zaban, A.; Gofer, Y.; Ely, Y. E.; Weissman, I.; Chusid, O.; Abramson, O. Recent studies of the lithium-liquid electrolyte interface electrochemical, morphological and spectral studies of a few important systems. *J. Power Sources* **1995**, *54* (1), 76–84.

(31) Rauh, R.; Brummer, S. The effect of additives on lithium cycling in propylene carbonate. *Electrochim. Acta* **1977**, *22* (1), 75–83.

(32) Ding, F.; Xu, W.; Graff, G. L.; Zhang, J.; Sushko, M. L.; Chen, X.; Shao, Y.; Engelhard, M. H.; Nie, Z.; Xiao, J.; et al. Dendrite-free lithium deposition via self-healing electrostatic shield mechanism. *J. Am. Chem. Soc.* **2013**, *135* (11), 4450–4456.

(33) Ota, H.; Shima, K.; Ue, M.; Yamaki, J.-i. Effect of vinylene carbonate as additive to electrolyte for lithium metal anode. *Electrochim. Acta* **2004**, *49* (4), 565–572.

(34) Mogi, R.; Inaba, M.; Jeong, S.-K.; Iriyama, Y.; Abe, T.; Ogumi, Z. Effects of some organic additives on lithium deposition in propylene carbonate. *J. Electrochem. Soc.* **2002**, *149* (12), A1578.

(35) Han, B.; Zhang, Z.; Zou, Y.; Xu, K.; Xu, G.; Wang, H.; Meng, H.; Deng, Y.; Li, J.; Gu, M. Poor Stability of Li<sub>2</sub>CO<sub>3</sub> in the Solid Electrolyte Interphase of a Lithium-Metal Anode Revealed by Cryo-Electron Microscopy. *Adv. Mater.* **2021**, *33* (22), 2100404.

(36) Wan, C.; Hu, M. Y.; Borodin, O.; Qian, J.; Qin, Z.; Zhang, J.-G.; Hu, J. Z. Natural abundance <sup>17</sup>O, <sup>6</sup>Li NMR and molecular modeling studies of the solvation structures of lithium bis (fluorosulfonyl) imide/1, 2-dimethoxyethane liquid electrolytes. *J. Power Sources* **2016**, *307*, 231–243.

(37) Yamada, Y.; Yaegashi, M.; Abe, T.; Yamada, A. A super-concentrated ether electrolyte for fast-charging Li-ion batteries. *Chem. Commun.* **2013**, *49* (95), 11194–11196.

(38) Zhang, G.; Deng, X.; Li, J.; Wang, J.; Shi, G.; Yang, Y.; Chang, J.; Yu, K.; Chi, S.-S.; Wang, H.; et al. A bifunctional fluorinated ether co-solvent for dendrite-free and long-term lithium metal batteries. *Nano Energy* **2022**, *95*, 107014.

(39) Hobold, G. M.; Khurram, A.; Gallant, B. M. Operando gas monitoring of solid electrolyte interphase reactions on lithium. *Chem. Mater.* **2020**, *32* (6), 2341–2352.

(40) Zhang, X. Q.; Cheng, X. B.; Chen, X.; Yan, C.; Zhang, Q. Fluoroethylene carbonate additives to render uniform Li deposits in lithium metal batteries. *Adv. Funct. Mater.* **2017**, *27* (10), 1605989.

(41) Su, C. C.; He, M.; Shi, J.; Amine, R.; Zhang, J.; Amine, K. Solvation Rule for Solid-Electrolyte Interphase Enabler in Lithium-Metal Batteries. *Angew. Chem.* **2020**, *132* (41), 18386–18390.

(42) Amanchukwu, C. V.; Kong, X.; Qin, J.; Cui, Y.; Bao, Z. Nonpolar alkanes modify lithium-ion solvation for improved lithium deposition and stripping. *Adv. Energy Mater.* **2019**, *9* (41), 1902116.

(43) Perez Beltran, S.; Cao, X.; Zhang, J.-G.; Balbuena, P. B. Localized high concentration electrolytes for high voltage lithium–metal batteries: correlation between the electrolyte composition and its reductive/oxidative stability. *Chem. Mater.* **2020**, *32* (14), 5973–5984.

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